

Cross-Disciplinary Scientific Retrieval Ontology: Breadth-First Research Map and Machine- Readable Baseline

Executive summary

The initially “unspecified” topic is resolved by the uploaded research brief: the operative task is to construct the **retrieval language of research itself**—first discovering the disciplinary structure of human research, then representing each field in its native terminology, while separately maintaining cross-domain, mathematical, computational-discovery, and quantum ontologies. The brief expressly rejects using AI, mathematics, computation, or quantum computing as relevance filters and prohibits premature cross-disciplinary hypothesis generation. [filecite turn0file](#)

Three interpretations were considered:

Possible interpretation	What it would study	Fit to the supplied brief
Scholarly-information market/infrastructure analysis	Publishers, databases, search platforms, ontology vendors, commercial/open scholarly infrastructure	Low. Useful later for implementation, but it would optimize around products rather than the structure and vocabulary of research.
Academic literature review of scientific knowledge organization	Research on taxonomies, ontologies, controlled vocabularies, scientometrics, semantic search, and classification	Medium. These sources are essential evidence, but the brief explicitly says the task is not a literature review.
Technical feasibility study plus first-pass construction of a universal retrieval ontology	A disciplinary tree, native vocabularies, lexical expansions, ambiguity controls, models/ equations/methods, source provenance, and coverage gaps	Selected. This directly corresponds to the requested structured outputs and breadth- first process.

The principal architectural conclusion is that a single universal taxonomy should **not** be treated as the ontology. A practical system needs at least three layers. The first is a broad hierarchical backbone for field coverage. The second consists of native disciplinary vocabularies, many of which are graphs or polyhierarchies rather than trees. The third contains independent retrieval overlays for concepts whose terminology crosses disciplinary boundaries. This follows directly from how major authoritative systems are constructed: ANZSRC has a three-level Fields of Research hierarchy; Gene Ontology allows multiple parents and typed relationships; APS PhySH organizes physics concepts simultaneously through facets and disciplines; and the Unified Astronomy Thesaurus is explicitly a SKOS thesaurus/polyhierarchy. ¹

A machine-readable **v0.1 breadth-first release** was constructed during this research. It contains:

Record family	Records in v0.1	Role
DISCIPLINES	2,215	Full ANZSRC FoR tree plus root and explicit independent retrieval overlays
CONCEPTS	2,706	One retrieval anchor for every ANZSRC leaf plus native and independent ontology concepts
TERMS	7,149	Search phrases, synonyms, aliases, abbreviations, broader terms, disambiguators, exclusions, high-recall expansions, historical terms
MODELS_EQUATIONS_METHODS	81	Named equations, algorithms, models, experimental paradigms, laws, transformations, and methods
RELATIONSHIPS	30	Conservative, explicitly encoded concept relations
SEED_SOURCES	34	Priority classification, thesaurus, ontology, standard, and terminology sources
COVERAGE	29	Coverage status and continuation gaps

The disciplinary backbone reproduces all **23 ANZSRC Divisions, 213 Groups, and 1,967 Fields** extracted from the current official FoR data cube; ANZSRC defines FoR as a Division → Group → Field hierarchy and publishes an explicit mapping to OECD FORD. The ABS made minor label and orthographic corrections to the downloadable classification on October 24, 2025, which are reflected in the currently published data cube used here. ²

This does **not** mean the ontology is complete. The hierarchy layer is comparatively strong; the native concept, terminology, historical-language, instrument, equation, and method layers remain deliberately marked **partial** or **shallow**. That distinction is essential: resource limits are not evidence of ontology completeness.

Interpretation and assumptions

The chosen interpretation is therefore:

Design and instantiate the first breadth-first version of a cross-disciplinary scientific-literature retrieval ontology, using authoritative disciplinary classifications and controlled vocabularies as evidence, without performing cross-domain hypothesis generation.

Several assumptions were required.

First, “human-knowledge research tree” is interpreted as **research fields represented in the scholarly/research ecosystem**, not literally every topic in all human knowledge. This includes natural sciences, formal

sciences, engineering, medicine, social sciences, humanities, creative practice research, law, education, Indigenous studies, built-environment research, and related research communities. ANZSRC is unusually suitable as the initial backbone because it is granular, explicitly hierarchical, maintained as an official research classification, and linked to the international OECD FORD framework. OECD FORD alone is too coarse for the stated objective: its highest level consists of six broad fields—natural sciences; engineering and technology; medical and health sciences; agricultural and veterinary sciences; social sciences; and humanities and arts. ³

Second, the disciplinary tree is treated as a **coverage scaffold, not an assertion that every research community has exactly one parent**. This distinction matters. A subject such as biophysics, computational neuroscience, environmental economics, mathematical physics, or quantum information naturally intersects several communities. Native systems already reflect this: PhySH permits concepts to belong to multiple facets or disciplines, while GO is a graph whose concepts may have multiple parents. ⁴

Third, the requested independent ontologies—mathematics, computation/discovery, quantum, and cross-cutting mechanisms—are represented as **explicit overlays rather than hidden cross-links used to decide relevance**. Thus “optimization” can occur as a mathematics concept, a computer-science retrieval term, and an independent cross-domain concept without asserting that all uses are mathematically or scientifically equivalent.

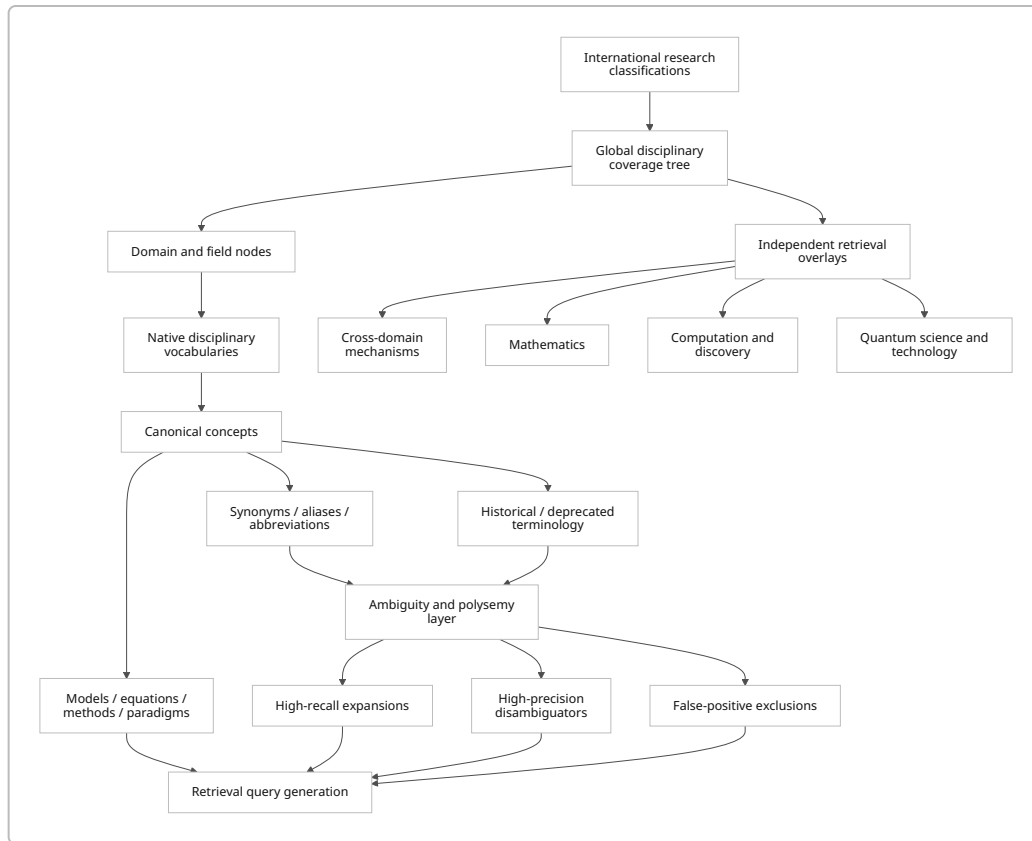
Fourth, English is the primary retrieval language in this release, as requested, but multilingual authoritative sources remain valuable. AGROVOC, for example, is a multilingual SKOS concept scheme and was last modified on March 3, 2026; CESSDA ELSST is explicitly multilingual for social-science terminology. Multilingual term ingestion is therefore a continuation layer rather than a reason to exclude such sources now. ⁵

Fifth, “most recent through August 14, 2026” is applied to **source-status verification**, not by replacing stable canonical classifications merely because they have older publication dates. MSC2020 remains the current Mathematics Subject Classification according to MathSciNet; MeSH 2026 is in production; the UAT website still identifies version 6.0.0 from December 16, 2025 as its current release even though its release announcement had expected 6.1.0 in spring 2026; and ISO/IEC 4879:2024 remains a published quantum-computing vocabulary standard. ⁶

A final assumption concerns licensing. “Authoritative source” does not automatically mean “safe to bulk-copy into a derivative ontology.” Open systems such as GO, UAT, and NERC NVS are designed for programmatic semantic use, while some standards and commercial/indexing vocabularies have stronger licensing restrictions. ISO’s page for ISO/IEC 4879:2024 explicitly states copyright restrictions on ISO materials, so a production ingestion pipeline must store **references and mappings separately from any term content whose license does not permit redistribution**. ⁷

Methodology and source architecture

The research used a **breadth-first source strategy**. The global backbone came first; domain-native terminology sources were then attached; independent overlays were built separately; only after that were ambiguity controls, named methods, and conservative relationships added.



The source hierarchy is deliberately heterogeneous because different disciplines have developed different forms of knowledge organization. The most useful sources are not necessarily all “ontologies” in the formal computer-science sense.

Domain	Priority source family	Why it is important for retrieval
Cross-disciplinary backbone	ANZSRC FoR + OECD FORD/Frascati	ANZSRC supplies detailed Division → Group → Field coverage and mappings to FORD; FORD supplies international high-level alignment. ⁸
Mathematics	MSC2020, AMS/zbMATH	Created and maintained by mathematical reviewing organizations specifically to classify mathematical literature; MathSciNet also exposes historical retired classes, which are valuable for retrospective retrieval. ⁹
Physics	APS PhySH	A regularly updated APS physics classification with research-area, physical-system, property, technique, and professional-topic facets. ¹⁰
Computing	ACM CCS + IEEE Thesaurus/Technology Navigator	ACM CCS supplies the canonical ACM computing ontology; IEEE's current navigator exposes more than 8,700 engineering/technology topics derived from the IEEE Thesaurus and says its topic taxonomy is updated annually. ¹¹

Domain	Priority source family	Why it is important for retrieval
Chemistry	IUPAC Gold Book and Color Books	IUPAC provides authoritative chemical terminology, nomenclature, symbols, and discipline-specific vocabulary; Gold Book v5.0.0 is the fifth edition/2025 online version. ¹²
Molecular/cellular biology	Gene Ontology	Formal species-agnostic biological concepts divided into Molecular Function, Cellular Component, and Biological Process, including synonyms, alternate IDs and typed relations. ¹³
Biomedicine and medicine	MeSH 2026	NLM's controlled hierarchical vocabulary for indexing and searching biomedical and health information; 2026 files are in production and available for download/API/RDF use. ¹⁴
Agriculture and food	FAO AGROVOC	Multilingual SKOS thesaurus with preferred and alternative terms, updated in 2026. ¹⁵
Earth/environment	NASA GCMD + GEMET	Complementary Earth-system and environmental vocabularies for science concepts, observations, platforms, instruments, and environmental indexing. ¹⁶
Oceanography	NERC Vocabulary Server / BODC P01	Standardized marine/ocean terms with IDs, abbreviations, definitions, version history, SKOS structures, APIs and SPARQL access. ¹⁷
Astronomy/astrophysics	Unified Astronomy Thesaurus	Open AAS-supported SKOS thesaurus; v6.0.0 contains 2,312 concepts, 11 top concepts and a depth of 11. ¹⁸
Economics	AEA JEL	Field-native classification used for economics literature; the AEA web classification tracks recent changes. ¹⁹
Broader social sciences	CESSDA ELSST	Broad social-science thesaurus spanning economics, sociology, politics, law, demography, health, employment and related areas. ²⁰
Education	ERIC Thesaurus	ERIC is the U.S. IES-sponsored education research database; its domain thesaurus supplies controlled education terminology. ²¹
Psychology	APA Thesaurus + Cognitive Atlas	APA supplies psychology index terminology; Cognitive Atlas separately connects cognitive constructs and experimental tasks. ²²
Linguistics/language	CLARIN vocabularies/CCR, plus specialized linguistic ontologies	Supplies semantic concepts used in linguistic annotation and metadata; the CLARIN ecosystem itself changed registry integration in 2026, demonstrating why versioning is necessary. ²³

Domain	Priority source family	Why it is important for retrieval
Materials science	MatPortal and materials ontologies	Aggregates materials-science ontologies; its June 2026 state reported dozens of ontologies and thousands of classes. ²⁴
Metrology	BIPM/JCGM VIM and GUM	International vocabulary and uncertainty framework for measurement; BIPM documented work toward VIM4, including a proposed revised uncertainty definition. ²⁵
Quantum computing	ISO/IEC 4879:2024 + NIST/physics vocabularies	ISO/IEC 4879 explicitly defines commonly used quantum-computing terms; physics vocabularies are still required for quantum science beyond computing. ²⁶
Robotics	IEEE 1872 series	CORA and its extensions supply robotics/automation ontological terminology. IEEE 1872-2015 became inactive-reserved on March 26, 2026 while a new P1872 revision is active; IEEE 1872.1-2024 and 1872.2-2021 remain active standards, and P1872.3 was approved as a project in June 2026. ²⁷

This evidence also determines the correct data model. A tree alone cannot preserve what native vocabularies know. GO, for example, distinguishes exact, broad, narrow and related synonyms, maintains obsolete identifiers, and allows multiple parent concepts; NERC NVS explicitly versions its controlled vocabularies and exposes accepted/deprecated concepts; MathSciNet preserves historical MSC classes. Therefore **source identifier, source version, preferred/non-preferred status, deprecation status, language, validity dates and provenance** should eventually become first-class fields, even though they were not part of the initially required table schemas. ²⁸

Disciplinary tree

The first deliverable is the breadth-first disciplinary scaffold. The official ANZSRC structure yields **23 Divisions, 213 Groups and 1,967 Fields**. A synthetic root plus eleven explicitly labelled independent overlay nodes produce the 2,215 **DISCIPLINES** records in v0.1. The FoR hierarchy and current downloadable data cube are official ABS data; ANZSRC also provides an explicit FoR-to-FORD correspondence. ²

The complete top-level distribution is:

Code	ANZSRC Division	Groups	Fields
30	Agricultural, Veterinary and Food Sciences	10	95
31	Biological Sciences	10	98
32	Biomedical and Clinical Sciences	16	129
33	Built Environment and Design	5	51
34	Chemical Sciences	8	63

Code	ANZSRC Division	Groups	Fields
35	Commerce, Management, Tourism and Services	10	91
36	Creative Arts and Writing	7	35
37	Earth Sciences	10	65
38	Economics	4	35
39	Education	5	43
40	Engineering	20	196
41	Environmental Sciences	7	38
42	Health Sciences	9	71
43	History, Heritage and Archaeology	4	44
44	Human Society	11	116
45	Indigenous Studies	20	320
46	Information and Computing Sciences	14	132
47	Language, Communication and Culture	6	91
48	Law and Legal Studies	8	62
49	Mathematical Sciences	6	48
50	Philosophy and Religious Studies	6	50
51	Physical Sciences	11	58
52	Psychology	6	36
Total		213	1,967

The distribution matters operationally: a seed list that says only “engineering,” “biology,” “computer science,” or “Indigenous studies” leaves vastly different numbers of leaf communities unrepresented. The breadth-first approach prevents research effort from being allocated according merely to familiarity. ANZSRC itself was revised partly to improve coverage of emerging research areas and was developed through consultation with researchers, organizations, professional associations, government bodies and Indigenous discipline experts. ²

ANZSRC field-level category counts by division

The chart above is calculated directly from the current official ANZSRC 2020 FoR data cube. ²

Several substantial branches appear that were absent from, or only indirectly represented in, the original seed list: **agricultural/veterinary/food sciences; built environment and design; education; law and legal studies; history/heritage/archaeology; Indigenous studies; creative arts and writing;**

commerce/management/tourism/services; and philosophy/religious studies. Their emergence is exactly why ontology completeness must be treated as a research problem rather than assumed from an initial list.

The hierarchy also demonstrates why familiar umbrella terms are inadequate. One example is:

Level	Identifier	Research area
Root	ROOT	Research literature
Division	ANZ-31	Biological Sciences
Group	ANZ-3103	Ecology
Field	ANZ-310301	Behavioural ecology

This particular Division → Group → Field example is also used by the ABS itself to illustrate FoR's hierarchy.

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Independent overlay branches were then added without altering the official classification:

Overlay ID	Overlay	Purpose
OV-CROSS	Cross-domain mechanisms	Search, inference, adaptation, feedback, criticality, transport, emergence, etc.
OV-COMPLEX	Complex systems and network science	Complex adaptive systems, networks, nonlinear dynamics, collective behavior
OV-MEAS	Scientific measurement and metrology	Measurement, uncertainty, calibration, traceability, instruments
OV-EXPDES	Experimental design and research methodology	Design, randomization, controls, power, replication, causal methodology
OV-SCISCI	Science of science and bibliometrics	Scientometrics, bibliometrics, research evaluation, scholarly communication
OV-MATH	Independent mathematics ontology	Mathematical objects, structures, operators, equations, proof and numerical methods
OV-DISCOVERY	Computation and scientific discovery	Program/algorithm discovery, automated reasoning, causal discovery, automated experimentation
OV-UNCONV	Unconventional and physical computation	Analog, optical, chemical, DNA, neuromorphic, reservoir and in-memory computation
OV-ALIFE	Artificial life	Digital evolution, self-reproduction, morphogenesis, open-ended evolution

Overlay ID	Overlay	Purpose
OV-QUANTUM	Quantum science and technology	Quantum phenomena, computation, communication, sensing, control, hardware and theory
OV-QBIO	Quantum biology	Field-native terminology for experimentally/theoretically studied quantum-biological phenomena

These are **retrieval overlays, not claims of equivalence or relevance**. For example, inclusion of quantum biology does not imply that any biological mechanism is useful for a quantum algorithm, and inclusion of “network” across multiple disciplines does not equate computer, neural, social, biological and mathematical networks.

Retrieval ontology and structured records

The package implements all seven requested record families. The excerpts below illustrate the populated representation; the downloadable files contain the complete v0.1 records.

DISCIPLINES

discipline_id	name	parent_id	level	description
ROOT	Research literature		0	Root of the retrieval hierarchy
ANZ-31	Biological Sciences	ROOT	1	Official ANZSRC Division
ANZ-3103	Ecology	ANZ-31	2	Official ANZSRC Group
ANZ-310301	Behavioural ecology	ANZ-3103	3	Official ANZSRC Field retrieval anchor
OV-QUANTUM	Quantum science and technology	ROOT	1	Independent quantum retrieval ontology
OV-QBIO	Quantum biology	OV-QUANTUM	2	Terminological quantum-biology overlay

CONCEPTS

concept_id	discipline_id	canonical_concept	concept_type	short_definition
CF-310202	ANZ-310202	Biological network analysis	other	ANZSRC-recognized research subject-area retrieval anchor

concept_id	discipline_id	canonical_concept	concept_type	short_definition
C000417	OV-CROSS	optimisation	method	Selection of solutions that extremize objectives subject to constraints
C000467	OV-CROSS	network	other	Polysemous retrieval token requiring field-qualified expansion
C000536	OV-DISCOVERY	automated theorem proving	computational_concept	Machine search for formal proofs
C000609	OV-QUANTUM	quantum error correction	framework	Encoding and recovery methods protecting quantum information from errors
C000612	OV-QUANTUM	surface code	model	Two-dimensional local-check topological stabilizer-code family
C000637	OV-MEAS	measurement uncertainty	other	Uncertainty associated with a measurement result

The biological representation should eventually be substantially enlarged from native sources such as GO. GO is particularly useful as a model because it stores canonical terms, definitions, IDs, alternate IDs, exact/broad/narrow/related synonyms and obsolescence status, rather than assuming that a canonical label is sufficient for retrieval. ¹³

TERMS

A core design goal is separating a concept from the strings used to retrieve it.

concept_id	term	term_type	context
C000469	attention	ambiguous	High-risk cross-domain polyseme

concept_id	term	term_type	context
C000469	selective attention \ visual attention \ attentional control \ attention mechanism \ self-attention	disambiguator	Field-qualified senses
C000469	news attention \ public attention	exclusion	Possible false-positive neighborhoods
C000469	"attention" AND (selective attention OR visual attention OR attentional control OR attention mechanism OR self-attention)	high_recall	Database-adaptable Boolean expansion
C000609	QEC	abbreviation	Quantum-error-correction shorthand
C000612	surface code	search_phrase	Exact model-family phrase

This is one of the most consequential parts of the ontology. Terms such as `state`, `network`, `field`, `phase`, `critical`, `attention`, `learning`, `representation`, `transport`, `control`, `signal`, `cell`, `population`, `fitness`, `stress`, `power`, `culture`, `agency`, and `risk` should generally **not be issued unqualified against a universal scholarly corpus**.

Representative ambiguity controls are:

Token	High-recall disciplinary senses	Typical false-positive problem
<code>network</code>	network science; graph; computer network; neural network; social network; biological network	Business networking, broadcast networks, unrelated network services
<code>state</code>	state variable; quantum state; mental state; disease state	Nation-state, U.S. state, government
<code>critical</code>	critical point; criticality; critical exponent; critical care; critical theory; critical thinking	The adjective “important/essential,” reviews or commentary
<code>attention</code>	selective/visual attention; attentional control; attention mechanism; self-attention	Public/media attention
<code>learning</code>	machine/statistical/reinforcement learning; human/animal/motor learning	Organizational-learning literature when unintended

Token	High-recall disciplinary senses	Typical false-positive problem
<code>representation</code>	knowledge, neural, mental, group, state-space representation	Political representation
<code>transport</code>	mass, charge, electron, membrane, public, optimal transport	Transportation-policy literature when physical transport is intended

The correct implementation is not a globally fixed exclusion list. It is a **concept × discipline × query-context matrix**. “Public transport” is a false positive for an electronic-transport query but a target term in transport engineering; “political representation” is a false positive for representation theory but a target in political science. This is why exclusions belong to contextual term records rather than to a universal stop-word list.

`MODELS_EQUATIONS_METHODS`

concept_id	name	type	discipline	related_concepts
<code>C000519</code>	Navier–Stokes equations	equation	mathematics / fluid dynamics	fluid dynamics; transport; nonlinearity
<code>C000681</code>	CRISPR–Cas	method	molecular biology / genetics	genome editing
<code>C000690</code>	Kalman filter	method	control / signal processing	state estimation
<code>C000697</code>	difference-in-differences	method	econometrics	causal inference
<code>C000718</code>	Schrödinger equation	equation	quantum mechanics	Hamiltonian; wave function
<code>C000724</code>	Shor’s algorithm	algorithm	quantum computing	integer factorization; period finding
<code>C000734</code>	Dijkstra’s algorithm	algorithm	graph algorithms	shortest path

The larger ontology should eventually treat these named entities as first-class retrieval objects. Searching only for “fluid dynamics,” for example, misses papers primarily described by `Navier–Stokes`, `RANS`, `LES`, `vorticity equation`, `finite-volume`, or specific turbulence-model names. The same principle applies throughout chemistry, molecular biology, economics, neuroscience, psychology, control theory, geophysics and quantum science.

`RELATIONSHIPS`

Relationships are intentionally sparse in v0.1 because the brief forbids manufactured analogies.

source_concept_id	relationship	target_concept_id	Interpretation
C000612 surface code	narrower_than	C000611 stabilizer code	Native quantum-code hierarchy
C000554 neural architecture search	narrower_than	C000533 algorithm discovery	Computational-search hierarchy
C000536 automated theorem proving	narrower_than	C000535 automated reasoning	Formal-reasoning hierarchy
C000539 symbolic regression	narrower_than	C000540 equation discovery	Discovery-method hierarchy
digital quantum simulation	narrower_than	quantum simulation	Quantum-simulation taxonomy
quantum key distribution	narrower_than	quantum communication	Quantum-communication taxonomy

No relationship of the form “biological adaptation = optimization,” “neural attention = transformer attention,” “phase transition = learning transition,” or similar has been manufactured. Such possible alignments belong to the later analytical stage, not this retrieval layer.

SEED_SOURCES

discipline_or_concept	source	year/status	role
Cross-disciplinary	ANZSRC Fields of Research	2020; data corrected 2025	Detailed disciplinary backbone 2
Mathematics	MSC2020	current	Native mathematical literature classification and historical classes 9
Physics	APS PhySH	current	Physics concepts, facets, disciplines 10
Biology	Gene Ontology	continuously revised	Concepts, relations, synonym scopes and obsolescence 13
Medicine	MeSH	2026	Biomedical indexing and retrieval vocabulary 29
Oceanography	NERC Vocabulary Server	current	Versioned SKOS vocabularies and parameter terminology 30
Astronomy	UAT 6.0.0	current release shown Aug. 2026	Astronomy/astrophysics polyhierarchical thesaurus 31

discipline_or_concept	source	year/status	role
Quantum computing	ISO/IEC 4879	2024, published	Standard quantum-computing vocabulary ³²
Robotics	IEEE 1872 family	active/revising in 2026	Robotics ontology standards ³³

COVERAGE

The most important semantic property of this table is that **partial** **never means “good enough to stop.”**

discipline/layer	status	Important missing work
Most ANZSRC scientific divisions	partial	Full native taxonomy ingestion, historical terms, instrument names, method vocabularies
Creative Arts and Writing	shallow	Native artistic-practice taxonomies, genres, methods, media, historical terminology
History/Heritage/Archaeology	shallow	Period/place-dependent terminology, archaeological methods and controlled heritage vocabularies
Indigenous Studies	shallow	Nation/community-specific terminology and governance-sensitive vocabularies; must avoid universalizing regional categories
Law and Legal Studies	shallow	Jurisdiction-specific vocabularies, legal doctrines, historical terminology, case-law indexing language
Philosophy and Religious Studies	shallow	Deeper PhilPapers/community taxonomies, traditions, named arguments, schools and historical terminology
Cross-domain mechanisms	partial	Exhaustive field-conditioned senses and exclusions
Independent mathematics	partial	Full MSC2020 ingestion, including its thousands of lower-level classes
Computation/discovery	partial	Full ACM/IEEE/formal-reasoning/program-synthesis terminology
Quantum	partial	Full standards + physics terminology, hardware vocabularies, code families, control terminology
Science of science	shallow	Indicators, bibliometric methods, research-policy and scholarly-communication vocabularies

MSC2020 alone illustrates the scale still ahead: it is a three-level literature classification maintained by MathSciNet and zbMATH, and external documentation of the 2020 revision reports 63 two-digit, 529 three-digit and 6,006 five-digit classifications. The current package therefore should not be mistaken for having

exhausted mathematical terminology merely because mathematics already has dozens of seeded concepts.

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Findings, implications, recommendations, and open questions

The strongest finding is that **ontology completeness and retrieval completeness are different problems**. A disciplinary tree can tell a system that oceanography exists; it cannot by itself retrieve thermohaline circulation, Ekman transport, CTD, ADCP, mixed layer depth, sea-surface salinity, instrument-model terminology, parameter codes, deprecated names, and regional vocabulary. The NERC Vocabulary Server demonstrates the retrieval-oriented alternative: standardized terms have persistent identifiers, abbreviations and definitions; vocabularies are versioned; accepted and deprecated concepts can be distinguished; and services expose them programmatically. 30

A second finding is that **native vocabulary structures must be preserved rather than flattened prematurely**. GO explicitly distinguishes exact, broad, narrow and related synonyms, retains alternate IDs after merges, supports obsolescence and replacement guidance, and lets terms have several parents. This maps almost directly onto the requested TERMS and RELATIONSHIPS requirements and suggests adding explicit provenance/version columns rather than storing only strings. 13

A third finding is that **historical terminology is not optional for massive-corpus retrieval**. MathSciNet exposes retired MSC classes spanning multiple historical periods; GO retains obsolete IDs; MeSH's annual updates can rename or discontinue publication types while keeping historical indexing searchable. A retrieval engine focused only on the current preferred label will systematically undersample older scientific literature. 35

A fourth finding is that source freshness has to be tracked **per vocabulary, not per project**. MeSH uses annual production vintages; AGROVOC exposes modification dates; GO is continuously revised; NVS is versioned; UAT releases numbered versions; and engineering/robotics standards have explicit active, inactive and revision states. A single ontology version field is therefore insufficient. 36

A fifth finding is that the original seed list was impressively broad but still exhibited a predictable scientific-technology bias. An independent official classification surfaced major research communities that would otherwise have received insufficient coverage, especially education, law, history/heritage/archaeology, built environment/design, Indigenous studies, creative arts/writing, agriculture/veterinary/food sciences and commerce/management/tourism/services. This validates the brief's instruction to make "ontology completeness itself a research problem." The ANZSRC review was explicitly intended to improve coverage and capture emerging research areas. 2

The prioritized continuation strategy should therefore be:

Priority	Action	Why
P0	Machine-ingest complete native taxonomies rather than manually adding famous terms	This is the largest recall gain and reduces familiarity bias. Begin with MSC2020, PhySH, GO/OBO, MeSH, AGROVOC, GCMD, NVS, UAT, ELSST/JEL and open computing vocabularies.
P0	Extend schema with <code>source_vocab</code> , <code>source_concept_id</code> , <code>source_version</code> , <code>language</code> , <code>preferred_status</code> , <code>deprecated</code> , <code>valid_from</code> , <code>valid_to</code> , <code>license</code> , <code>provenance</code>	Required to preserve versioned, historical and polyhierarchical source information.
P0	Build an explicit licensing ledger before bulk ingestion	Some authoritative standards/indexing vocabularies cannot simply be redistributed; open and restricted sources need different ingestion/mapping strategies. ISO's own page is a concrete warning. ³²
P1	Construct field-conditioned ambiguity templates	This directly addresses high-frequency tokens such as <code>state</code> , <code>field</code> , <code>network</code> , <code>attention</code> , <code>critical</code> , <code>control</code> , and <code>representation</code> .
P1	Add named-equation/model/method dictionaries field by field	Many research communities retrieve more effectively through named models, instruments and methods than through broad subject labels.
P1	Ingest historical/deprecated terminology and version crosswalks	Essential for retrospective corpora and century-scale research retrieval.
P1	Add instruments, observables, assay names, experimental paradigms and dataset terminology	Particularly important in astronomy, Earth science, chemistry, medicine, neuroscience, oceanography and engineering.
P2	Add multilingual aliases after English coverage stabilizes	Improves global literature recall without allowing translation work to block English breadth-first completion.
P2	Run corpus-driven "unknown vocabulary harvesting"	Sample papers from every leaf field, identify high-frequency noun phrases not mapped to ontology concepts, and recursively create candidate terms for expert/source validation.

Priority	Action	Why
P2	Measure coverage empirically	For each discipline, test recall against stratified known-paper sets, historical corpora, journals and database subject headings rather than declaring completeness by record count.

The most important methodological recommendation is the last one. **No finite authoritative taxonomy is a proof of retrieval completeness.** A production system should alternate between top-down ontology ingestion and bottom-up corpus discovery: retrieve literature using known terms, mine unexplained field-specific terminology, trace those terms back to native authoritative sources, add validated records, and repeat.

Several open questions remain unresolved rather than silently assumed:

Open question	Why it matters
What scholarly databases will ultimately be queried —Crossref, OpenAlex, PubMed, arXiv, Semantic Scholar, Web of Science, Scopus, discipline indexes, publisher search, full-text corpora?	Boolean syntax, field operators, stemming, controlled headings, phrase handling and exclusion behavior differ by retrieval target.
Which licensed vocabularies may legally be copied versus only referenced or mapped?	Determines whether the corpus ontology can redistribute labels or must store external identifiers and query mappings.
How should a concept with multiple legitimate parents be represented in <code>DISCIPLINES</code> ?	A strict parent field cannot reproduce polyhierarchies; either the tree remains a coverage scaffold or the schema must permit multiple parent edges.
How should granularity be normalized?	One vocabulary may classify a topic as a field, another as a method, and another as a narrow concept. Flattening these distinctions would lose disciplinary meaning.
How will ontology quality be evaluated without a comprehensive gold standard?	Recall tests, journal stratification, source-vocabulary coverage and “unmapped terminology rate” are likely more informative than raw record counts.
How should eponymous and historical terminology be handled as naming conventions evolve?	Retrospective search requires preserving historical labels while current retrieval may prefer modern, non-eponymous or standardized terminology.

Open question	Why it matters
How should regionally governed Indigenous terminology be represented?	A global ontology must not collapse community-, nation-, country- or language-specific concepts into an invented universal hierarchy.
When should cross-domain semantic alignment occur?	Only after retrieval coverage is substantially stronger; doing it now would risk encoding the very hypothesis bias the brief prohibits.

Accordingly, the correct next stage is **not** “discover surprising connections.” It is continued ontology acquisition, normalization, provenance recording, historical-language capture, and coverage testing.

Prioritized bibliography and deliverables

The following bibliography is prioritized by direct utility to a universal retrieval system rather than by academic prestige.

Priority	Source	Retrieval role
A	Australian Bureau of Statistics, ANZSRC 2020 Fields of Research. ²	Main breadth-first disciplinary scaffold; detailed definitions and FORD correspondence
A	OECD, Frascati Manual 2015 and FORD framework. ³⁷	International high-level R&D classification alignment
A	AMS/zbMATH, Mathematics Subject Classification MSC2020. ⁹	Native mathematics hierarchy and historical classification language
A	American Physical Society, Physics Subject Headings — PhySH. ¹⁰	Native physics concepts, techniques and facets
A	Gene Ontology Consortium, Gene Ontology. ¹³	Biology ontology architecture, relations, synonym scopes and deprecated terms
A	U.S. National Library of Medicine, MeSH 2026. ¹⁴	Biomedical controlled terms, hierarchy, entry terms and annual updates
A	IUPAC, Compendium of Chemical Terminology / Gold Book, 5th ed. ¹²	Canonical chemical terminology and definitions
A	FAO, AGROVOC Multilingual Thesaurus. ¹⁵	Agriculture, food, forestry and related terminology
A	British Oceanographic Data Centre, NERC Vocabulary Server and P01 Parameter Usage Vocabulary. ¹⁷	Marine variables, parameters, abbreviations, persistent IDs and version history

Priority	Source	Retrieval role
A	American Astronomical Society, Unified Astronomy Thesaurus v6.0.0. ³⁸	Astronomy and astrophysics concepts, synonyms and semantic relationships
A	ISO/IEC, 4879:2024 Information technology — Quantum computing — Vocabulary. ³²	Standardized quantum-computing terminology; licensing must be respected
A	IEEE, 1872 robotics/automation ontology family. ³³	Robotics, autonomy and robot-task terminology
B	ACM, Computing Classification System 2012. ³⁹	Computing subject classification
B	IEEE, Technology Navigator / IEEE Thesaurus. ⁴⁰	Broad current engineering and technology terminology
B	NASA, GCMD Keywords. ⁴¹	Earth science, instruments, platforms and observational metadata
B	EEA/Eionet, GEMET. ⁴²	Environmental terminology
B	CESSDA, ELSST v6. ²⁰	Multilingual social-science retrieval vocabulary
B	American Economic Association, JEL Classification System. ¹⁹	Economics literature taxonomy
B	Institute of Education Sciences, ERIC and ERIC thesaurus. ²¹	Education research vocabulary
B	APA, Thesaurus of Psychological Index Terms. ⁴³	Psychology search/indexing terms and preferred/non-preferred vocabulary
B	Cognitive Atlas. ⁴⁴	Cognitive concepts and experimental paradigms/tasks
B	CLARIN, Concept Registry / vocabulary infrastructure. ²³	Linguistic annotation and metadata terminology
B	BIPM/JCGM, International Vocabulary of Metrology and VIM4 development materials. ⁴⁵	Measurement, uncertainty, calibration and metrological language
C	MatPortal and materials-science ontology ecosystem. ²⁴	Materials terminology and ontology discovery
C	PhilPapers category system. ⁴⁶	Community-maintained philosophy subject structure

The complete machine-readable v0.1 package is available here:

[Download the complete scientific retrieval ontology v0.1 ZIP](#)

Individual primary outputs:

File	Download
DISCIPLINES	DISCIPLINES.csv
CONCEPTS	CONCEPTS.csv
TERMS	TERMS.csv
MODELS_EQUATIONS_METHODS	MODELS EQUATIONS METHODS.csv
RELATIONSHIPS	RELATIONSHIPS.csv
SEED_SOURCES	SEED_SOURCES.csv
COVERAGE	COVERAGE.csv
Quantitative backbone counts	ANZSRC_DIVISION_COUNTS.csv
Release notes	README.md

Equivalent `.jsonl` representations are also included in the ZIP for programmatic ingestion.

The principal continuation point is unambiguous: **preserve this complete breadth-first scaffold, then recursively ingest authoritative native vocabularies into it—especially their synonyms, deprecated/historical terms, model and method names, instrument language, identifiers, semantic relations and version metadata—while keeping all cross-domain alignment analytically separate.** That approach maximizes the probability that future computational research can retrieve literature expressed in terminology its designers did not already know to search for.

¹ ² ⁸ <https://www.abs.gov.au/statistics/classifications/australian-and-new-zealand-standard-research-classification-anzsrc/2020>

<https://www.abs.gov.au/statistics/classifications/australian-and-new-zealand-standard-research-classification-anzsrc/2020>

³ <https://uis.unesco.org/en/glossary-term/fields-research-and-development-ford>

<https://uis.unesco.org/en/glossary-term/fields-research-and-development-ford>

⁴ ¹⁰ <https://physh.org/>

<https://physh.org/>

⁵ ¹⁵ <https://agrovoc.fao.org/browse/agrovoc/en/index/0-9>

<https://agrovoc.fao.org/browse/agrovoc/en/index/0-9>

⁶ ⁹ ³⁴ ³⁵ <https://mathscinet.ams.org/mathscinet/info/docs/search-extras/search-msc>

<https://mathscinet.ams.org/mathscinet/info/docs/search-extras/search-msc>

⁷ ¹³ ²⁸ <https://www.geneontology.org/docs/ontology-documentation/>

<https://www.geneontology.org/docs/ontology-documentation/>

- 11 39 <https://ccecc.acm.org/guidance>
<https://ccecc.acm.org/guidance>
- 12 <https://mail.goldbook.iupac.org/terms/view/09693/plain>
<https://mail.goldbook.iupac.org/terms/view/09693/plain>
- 14 29 36 <https://sites.wip.nlm.nih.gov/mesh/meshhome.html>
<https://sites.wip.nlm.nih.gov/mesh/meshhome.html>
- 16 41 <https://gcmd.earthdata.nasa.gov/static/kms/>
<https://gcmd.earthdata.nasa.gov/static/kms/>
- 17 30 https://www.bodc.ac.uk/resources/products/web_services/vocab/
https://www.bodc.ac.uk/resources/products/web_services/vocab/
- 18 <https://astrothesaurus.org/release-of-uat-version-6-0-0/>
<https://astrothesaurus.org/release-of-uat-version-6-0-0/>
- 19 <https://www.aeaweb.org/econlit/jelCodes.php?print=&view=jel>
<https://www.aeaweb.org/econlit/jelCodes.php?print=&view=jel>
- 20 <https://elsst.cessda.eu/about/>
<https://elsst.cessda.eu/about/>
- 21 <https://ies.ed.gov/use-work/education-research-database-eric>
<https://ies.ed.gov/use-work/education-research-database-eric>
- 22 43 <https://www.apa.org/pubs/books/3100084.html>
<https://www.apa.org/pubs/books/3100084.html>
- 23 <https://dev.clarin.nl/node/4216.html>
<https://dev.clarin.nl/node/4216.html>
- 24 <https://matportal.org/>
<https://matportal.org/>
- 25 <https://www.bipm.org/en/-/2025-07-03-webinar-explores-future-definition-of-measurement-uncertainty-1>
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- 26 32 <https://www.iso.org/standard/80432.html>
<https://www.iso.org/standard/80432.html>
- 27 33 <https://standards.ieee.org/ieee/1872/12250/>
<https://standards.ieee.org/ieee/1872/12250/>
- 31 38 <https://astrothesaurus.org/>
<https://astrothesaurus.org/>
- 37 https://www.oecd.org/en/publications/frascati-manual-2015_9789264239012-en.html
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- 40 <https://technav.ieee.org/>
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- 42 <https://www.eionet.europa.eu/gemet/en/about/>
<https://www.eionet.europa.eu/gemet/en/about/>

⁴⁴ <https://www.cognitiveatlas.org/>

<https://www.cognitiveatlas.org/>

⁴⁵ <https://www.bipm.org/en/doi/10.59161/jcgm200-2012>

<https://www.bipm.org/en/doi/10.59161/jcgm200-2012>

⁴⁶ <https://philpapers.org/>

<https://philpapers.org/>